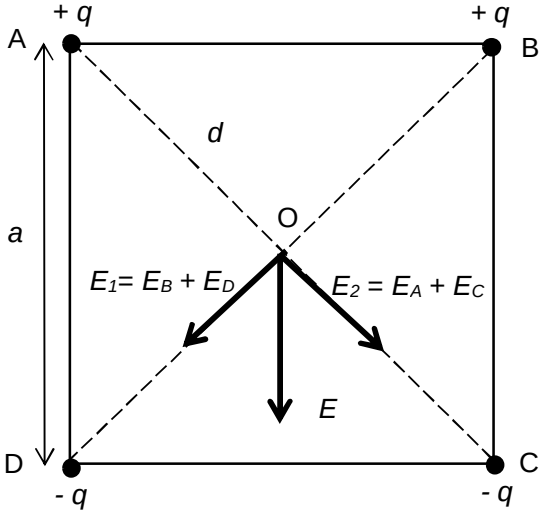


5(a)	Electric field strength at a point is the electric force per unit positive charge of a small test charge placed at that point. <u>Common Mistake:</u> <i>Electric field strength at a point is the electric force per unit positive charge acting on that point. (electric force acts on the charge, not the point)</i>	[1]
5(b)(i) and (ii)	 Distance $AO = BO = CO = DO = d$, where $d = \frac{1}{2} \sqrt{a^2 + a^2} = \frac{a}{\sqrt{2}}$ E_A, E_B, E_C and E_D are the electric field strength due to charges at A, B, C and D respectively. $E_A = E_B = E_C = E_D = \frac{q}{4\pi\epsilon_0 \left(\frac{a}{\sqrt{2}}\right)^2} = \frac{2q}{4\pi\epsilon_0 a^2}$ E_B and E_D are in the same direction while E_A and E_C are in the same direction. Thus $E_1 = E_B + E_D$ and $E_2 = E_A + E_C$ $E_1 = E_2 = \frac{4q}{4\pi\epsilon_0 a^2} = \frac{q}{\pi\epsilon_0 a^2}$ The resultant electric field strength at point O is $E = \sqrt{E_1^2 + E_2^2} = \frac{\sqrt{2}q}{\pi\epsilon_0 a^2}$	[1] draw E correctly [1] for $\frac{a}{\sqrt{2}}$ [1] for $\frac{2q}{4\pi\epsilon_0 a^2}$ [1]
(c)	The particle accelerates from AB towards CD along a straight path. Upon exiting CD, it decelerates.	[1] [1]

	<u>Common Mistake:</u> Described the forces acting on the charge instead of describing the motion in terms of its acceleration and direction.	
(d)	Potential at C is $V_C = V_A + V_B + V_D$ $= \frac{1}{4\pi\epsilon_0} \left(\frac{q}{\sqrt{2}a} + \frac{q}{a} - \frac{q}{a} \right)$ $= \frac{q}{4\sqrt{2}\pi\epsilon_0 a}$ Minimum KE = change in p.e. as charge moves to infinity $= qV_C = \frac{q^2}{4\sqrt{2}\pi\epsilon_0 a}$	[1] for potential at C [1] for conservation of energy [1] for correct expression
6(a)(i)	From Fig. 6.2. there is constant rate of decrease of magnetic flux	[1] for correct